

Practical No. 10

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Class : SY-B

Branch – AI & DS

Roll no : B-12

Aim - Version Control using Git & GitHub Questions :

1. What is the difference between Git and GitHub? Ans

: Git vs GitHub

- Git is a Version Control System (VCS)
- Created by Linus Torvalds in 2005
- Works completely offline on your local machine
- Tracks every change made to your code
- Stores full project history locally
- Free and open-source
- Used via command line (terminal)
- Alternatives: Mercurial, SVN
- GitHub is a cloud-based hosting platform for Git repos
- Founded in 2008, bought by Microsoft in 2018
- Requires internet connection
- Adds collaboration features: Pull Requests, Issues, Actions
- Stores your code remotely on servers
- Has free and paid plans
- Alternatives: GitLab, Bitbucket

2. Explain the difference between:

- git add

- git commit

Ans : git add:

- Moves files from Working Directory → Staging Area
- Tells Git which changes to include in next commit
- Does NOT save permanently
- Can stage one file, multiple files, or all files
- Reversible using `git restore --staged`

Commands:

`Git add file.txt` → one file

`Git add .` → all files

`Git add *.js` → all JS files

`Git add -p` → select parts interactively

commit:

- Moves files from Staging Area → Local Repository
- Creates a permanent snapshot in history
- Each commit gets a unique SHA-1 hash ID
- Requires a commit message • Records author, timestamp, and changes

`Git commit -m "message"` → basic commit

`Git commit --amend` → edit last commit

`Git commit -am "message"` → add + commit together

3.What is the staging area in Git?

Ans : • Also called the Index

- It is a middle layer between working directory and repository
- Physically stored in `.git/index` file
- Lets you selectively choose what to commit
- You can modify 10 files but commit only 2
- Helps create clean, logical commits • You can review staged changes before saving

- Commands:

Git status → see staged/unstaged files

Git diff –staged → see staged changes Git

restore –staged file → unstage a file -

4 States of a file:

- Untracked → new file Git doesn't know about
- Modified → tracked file with new changes
- Staged → ready to be committed
- Committed → permanently saved in history

4. What happens if you modify a file after committing it?

Ans : • The old committed version stays safe in history

- Modified file appears as “modified” in working directory
- Git detects change by comparing file hash with last commit
- Git status shows it under “Changes not staged for commit”
- You must add and commit again to save new version
- Each commit gets a new unique hash
- Old commit is still accessible anytime
- You can go back using git checkout <old-hash>
- Nothing in Git is truly lost after committing
- To fix last commit quickly: git commit –amend Process after modifying:

Modify file → git add → git commit → new snapshot created

5. Difference between:

- git push

- git pull

Ans : git push:

- Uploads local commits → remote repository (GitHub)
- Shares your work with teammates
- Only pushes committed changes (not just staged)

- May be rejected if remote has newer commits Commands:
 - Git push origin main → push to main branch
 - Git push origin branch-name → push specific branch
 - Git push -u origin main → set default upstream
 - Git push -force → force overwrite (dangerous) Git pull:

- Downloads remote changes → local machine
- Keeps your local repo up to date
- Combines: git fetch + git merge
- Can cause merge conflicts if same lines were edited Commands:
 - Git pull origin main → pull from main
 - Git pull -rebase → pull and rebase instead of merge

6 .What is Version Control? Differentiate between Centralized v/s Distributed VCS.

Ans : Version Control:

- System that records changes to files over time
- Tracks who, what, when, and why something changed
- Allows reverting to any previous version
- Enables parallel work without overwriting each other
- Essential for team software development Centralized VCS (CVCS):
 - One central server holds the entire repository
 - Developers check out only one version (not full history)
 - All operations need server/internet connection
 - Examples: SVN, CVS, Perforce - Advantages:
 - Simple to understand
 - Admin has full control
 - Good for binary/design files Distributed VCS (DVCS):
 - Every developer has full copy of entire repo including history

- Most operations happen locally (fast)
- Remote server used only for syncing/sharing
- Examples: Git, Mercurial - Advantages:
- Fully offline capability
- Every clone = complete backup
- Very fast operations
- Easy branching and merging
- No single point of failure
- **Theroy :**

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository.](#)
 Required fields are marked with an asterisk (*)

1 General

Owner * NageshNaikwade / Repository name * PDS-Practicals
 ⚠ PDS-Practicals already exists in this account

Great repository names are short and memorable. How about [stunning-sniffle](#)?

Description
 0 / 350 characters

2 Configuration

Choose visibility *
 Choose who can see and commit to this repository Public

Add README
 READMEs can be used as longer descriptions. [About READMEs](#) Off

Add .gitignore
 .gitignore tells git which files not to track. [About ignoring files](#) No .gitignore

Add license No license

```

lenovoLAPTOP-80NSF3M5 MINGW64 ~/myproject/myproject1 (master)
$ git remote add origin http://github.com/NageshNaikwade/PDS-Practicals.git

lenovoLAPTOP-80NSF3M5 MINGW64 ~/myproject/myproject1 (master)
$ git push -u origin master
remote: Permission to NageshNaikwade/PDS-Practicals.git denied to abhays0777.
fatal: unable to access 'https://github.com/NageshNaikwade/PDS-Practicals.git/': The requested URL returned error: 403

lenovoLAPTOP-80NSF3M5 MINGW64 ~/myproject/myproject1 (master)
$ git remote add origin https://github.com/NageshNaikwade/PDS-Practicals.git
error: remote origin already exists.

lenovoLAPTOP-80NSF3M5 MINGW64 ~/myproject/myproject1 (master)
$ git remote add origin https://github.com/NageshNaikwade/PDS-Practicals.git
error: remote origin already exists.

lenovoLAPTOP-80NSF3M5 MINGW64 ~/myproject/myproject1 (master)
$ git push -u origin master
info: please complete authentication in your browser...
warning: redirecting to https://github.com/NageshNaikwade/PDS-Practicals.git/
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 208 bytes | 208.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:   https://github.com/NageshNaikwade/PDS-Practicals/pull/new/master
remote:
To http://github.com/NageshNaikwade/PDS-Practicals.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

lenovoLAPTOP-80NSF3M5 MINGW64 ~/myproject/myproject1 (master)
$

```

```

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject (master)
$ git --version
git version 2.34.0.windows.1

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject (master)
$ git config --global user.name "nagesh"

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject (master)
$ git config --global user.email "nageshnaikwade0@gmail.com"

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=nagesh
user.email=nageshnaikwade0@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true

```

```

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject (master)
$ mkdir myproject
mkdir: cannot create directory 'myproject': File exists

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject (master)
$ mkdir myproject1

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject (master)
$ cd myproject1

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject/myproject1 (master)
$ git init
Initialized empty Git repository in C:/Users/Lenovo/Myproject/myproject1/.git/

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject/myproject1 (master)
$ touch file.txt

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject/myproject1 (master)
$ git add file.txt

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject/myproject1 (master)
$ git add.
git: 'add.' is not a git command. See 'git --help'.

The most similar command is
    add

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject/myproject1 (master)
$ git add
Nothing specified, nothing added.
hint: Maybe you wanted to say 'git add .' ?
hint: Disable this message with 'git config set advice.addEmptyPathsSpec false'

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject/myproject1 (master)
$ git add
Nothing specified, nothing added.
hint: Maybe you wanted to say 'git add .' ?
hint: Disable this message with 'git config set advice.addEmptyPathsSpec false'

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject/myproject1 (master)
$ git add .

lenovo\laptop-BONGSFJMS MINGW64 ~/myproject/myproject1 (master)
$ git commit -m "first commit"
[master (root-commit) b7ed27c] first commit
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 file.txt

```

NageshNaikwade / PDS-Practicals

Type / to search

<> Code

Issues

Pull requests

Agents

Actions

Projects

Wiki

Security and quality

Insights

Settings

On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. [Review this update](#) and manage your preferences in your [GitHub account settings](#).

PDS-Practicals

Public

Pin

Watch 0

Fork 0

Star 0

master had recent pushes 23 seconds ago

Compare & pull request

master

2 Branches

0 Tags

Go to file

Add file

<> Code

This branch is 1 commit ahead of and 10 commits behind main.

Contribute

nagesh

first commit

b7ed27c · 13 minutes ago

1 Commit

file.txt

first commit

13 minutes ago

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 1

NageshNaikwade

Languages

Jupyter Notebook

100.0%